

Sam Shelly

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AI Engineer focused on locally-hosted, secure AI systems for privacy-sensitive environments

EDUCATION

Vassar College

Aug 2022 – May 2026

Bachelor of Arts in Computer Science, Minor in Astronomy

Poughkeepsie, NY

GPA: 3.64/4.00 | Major GPA: 3.68/4.00 | Coursework: Data Structures & Algorithms, Analysis of Algorithms, Software Design, Compilers, Theory of Computation, Computer Organization

EXPERIENCE

AI Development Intern

May 2025 – Jan 2026

Innovative Computing Systems

Los Angeles, CA

- Led a four person team to architect a Microsoft Copilot Studio knowledge agent for a 60-person systems integration firm.
- Restricted the agent to closed, company-fed data sources to protect client IP and eliminate public internet exposure.
- Built Python ingestion pipelines and ConnectWise API integrations to power AI-driven enterprise search, reducing information retrieval time by 30%.
- Launched a 15-user pilot and led technical evaluations for clients, advancing the product towards commercialization.

Computer Science Department Coach

Jan 2024 – May 2026

Vassar College

Poughkeepsie, NY

- Led three sessions per week on algorithms, data structures, and debugging in Java for up to 50 students per semester (200+ total), adapting pacing and session plans for struggling students.
- Mentored two coaches, and met weekly with the course professor to align on curriculum direction and student learning progress.

PROJECTS

AI Knowledge Base RAG

GitHub

Python, FastAPI, LangChain, Qdrant, Ollama, Mistral 7B, BAAI/bge-large-en-v1.5, PyMuPDF, Docker, Streamlit, pytest, GitHub Actions

- Built a closed-loop Retrieval-Augmented Generation (RAG) system using self-hosted, open-weight components with no external API calls to securely query government and defense AI policy documents in air-gapped and other privacy-sensitive environments.
- Developed a custom LLM-as-judge evaluation pipeline achieving 0.95 retrieval recall and 0.92 answer similarity across a ground-truth QA dataset, with results visualized in a Streamlit dashboard.
- Built an end-to-end document ingestion and retrieval pipeline using PyMuPDF, semantic chunking, BAAI/bge-large-en-v1.5 embeddings, Qdrant vector search, and Mistral 7B (Ollama) for grounded, cited responses.
- Deployed the application as a Dockerized FastAPI REST service with automated testing and CI/CD using GitHub Actions.

DataFest 2024 — Best Overall Analysis

48-hr competition

Python, pandas

- Analyzed student-interaction data (video views, quiz attempts, textbook page views; $n \approx 1,857$ students) from an online statistics textbook to advise an ed-tech client on improving learning outcomes.
- Built a multiple regression model ($R^2=0.47$) identifying quiz effort and video engagement as positive predictors of student performance, and quantified that textbook-only engagement and frequent answer-changing predicted lower scores.
- Built a pandas-based directed dependency graph mapping how students navigated between textbook chapters/sections while studying, informing a recommendation to prepend review material to high-traffic “capstone” chapters.
- Authored final write-up and slide presentation synthesizing both analyses into client-facing recommendations.

PUBLICATIONS

Feedback-Driven Magnetic Field Evolution in MW-type Simulated Circumgalactic Medium — KNAC Undergraduate Symposium, 2024 | **Efficacy of NER Models on Fiction and Nonfiction Domains** — CMPU 366 NLP, Vassar College, 2025

SKILLS

Languages: Python, Java, C, C++, C#, OCaml | **ML/AI:** PyTorch, HuggingFace Transformers, LangChain, Ollama, Qdrant | **Tools:** Docker, FastAPI, Streamlit, GitHub Actions, PyMuPDF, RISC-V, pytest